EEG-Based Motor Action and Motor Imagery Classification Using Machine Learning

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Abstract

This work presents an EEG-based classification framework for distinguishing motor actions and motor imagery using a consumer-grade Emotiv EPOC X headset. EEG recordings were collected from 47 participants using 14 channels sampled at 128 Hz while participants performed or imagined six limb movements involving the right and left arms, legs, and wrists. The resulting protocol formed twelve task classes by combining the body part with the action or imagery condition. The raw EDF recordings were processed using MNE-Python, including channel selection, 4–45 Hz band-pass filtering, signal-quality based trial rejection, per-recording normalisation and sliding-window segmentation. Each 10-second task segment was divided into 2-second windows with a 0.5-second stride, producing 8,092 windows from 476 trials. A heterogeneous representation of 1,771 features per window was then constructed using spectral, statistical, Hjorth, fractal and wavelet descriptors, together with spectrogram, scalogram, inter-hemispheric connectivity and Riemannian tangent-space features. An ensemble of Logistic Regression, Linear Discriminant Analysis, Extra Trees and Histogram Gradient Boosting was used as the main classifier, and Support Vector Machine, Random Forest, Naive Bayes, AdaBoost, EEGNet and CNN–LSTM models were evaluated for comparison. Subject-level GroupKFold validation was used to prevent windows from the same participant appearing in both training and evaluation folds. The ensemble achieved 80.25% for binary action-versus-imagery classification, 26.68% for six-class body-part classification and 21.43% for the full

twelve-class task, while left-versus-right imagery classification reached 52.28% under nested cross-validation. The results show that combining complementary time–frequency, covariance-based and signal-derived features provides a strong baseline for subject-independent decoding of real versus imagined movement from low-cost EEG, while limb-level and side-level decoding remain challenging.

Keywords: Electroencephalography (EEG), Brain–Computer Interface, Motor Imagery, Motor Action, Spectrogram, Scalogram, Riemannian Geometry, Machine Learning, Emotiv EPOC X

1 Introduction

Brain–computer interfaces (BCIs) give the brain a channel to a computer that does not pass through muscles. They record brain activity, look for patterns that match an intention, and turn those patterns into a command. Motor applications are where this matters most. A person who has lost control of a limb after a stroke or a spinal injury can often still produce the cortical activity that would normally drive the movement, and a BCI that reads this activity could steer a wheelchair, open a prosthetic hand or guide a rehabilitation robot. Most such systems use electroencephalography (EEG). It is non-invasive and cheap compared with other recording methods, it follows brain activity on the scale of milliseconds, and it can be recorded with equipment small enough to carry [1, 2].

Two related mental states are central to this work. In motor execution, or *action*, the person really makes the movement. In motor imagery, the same movement is rehearsed in the mind while the body stays still. The two are not the same, but they share much of their neural machinery. Both reduce the power of the sensorimotor rhythms over the motor cortex, mostly in the mu band (8–12 Hz) and the beta band (13–30 Hz), and this drop, known as event-related desynchronisation, is the signal that most motor-imagery BCIs rely on [3, 4].

Much of the early work kept the problem small. The classic task is to tell imagined left-hand movement from imagined right-hand movement, and Common Spatial Pattern (CSP) filters followed by linear discriminant analysis or a support vector machine became the standard way to solve it [5, 6]. Adding more classes makes things far harder. The cortical areas for the hand, the wrist and the leg lie close together, the skull spreads their activity across the scalp, and the EEG of one person can look quite unlike that of the next [7, 8]. A model that works well for the people it was trained on can therefore fail on someone new.

The recording hardware adds its own limits. Clinical EEG amplifiers give clean signals, but they are expensive, slow to set up and rarely leave the laboratory. Consumer headsets such as the Emotiv EPOC are cheap, quick to put on and comfortable enough for everyday use, and studies have shown that they capture real neural activity [9, 10]. They are also noisier, more sensitive to movement and limited in where they place their electrodes [11]. The EPOC X used here has fourteen saline electrodes,

none of which sits over the central motor strip (C3, Cz and C4), which is where motor activity is usually strongest.

Researchers have tried many ways of turning short EEG segments into something a classifier can use. Band-power and statistical features summarise each channel. Spectrograms and scalograms keep track of how the rhythms change over time. Riemannian methods describe a trial by the covariance between its channels and work directly with the geometry of these matrices [12, 13]. Deep networks such as EEGNet learn their own features from the raw signal [14, 15]. Each of these views captures something the others miss, so combining them is a natural step, particularly when the dataset is too small for a deep network to learn everything on its own.

How a model is evaluated matters as much as the model itself. EEG windows cut from the same recording are strongly correlated, so if windows of one person appear in both the training and the test set, the classifier can learn to recognise the person instead of the task. An early version of our own pipeline fell into exactly this trap and reported accuracies above 90%, which dropped sharply once the evaluation was corrected. For a BCI that has to work for new users, the only fair test is one in which the people being tested were never seen during training.

In this study we built a complete classification pipeline around the Emotiv EPOC X and a twelve-task protocol. Participants performed and then imagined six movements: right-arm and left-arm flexion, right-leg and left-leg lifting, and right-wrist and left-wrist rotation. We analysed the same recordings at three levels of detail: action versus imagery (two classes), the body part involved (six classes), and the exact task (twelve classes). At our guide's suggestion we also asked whether the side of an imagined movement, left or right, can be decoded. Along the way we found that the task schedule did not start at a fixed time in every recording, and we recovered the true start from the headset's own motion sensor before cutting any trials. Every result we report comes from people the model had not seen during training.

The main contributions of this work are as follows:

1. **A new twelve-task EEG dataset of real and imagined limb movements.** We recorded 47 participants with a consumer-grade 14-channel Emotiv EPOC X while each of six movements (right and left arm, leg and wrist) was first performed and then imagined.
2. **Subject-independent decoding of executed versus imagined movement from consumer-grade EEG.** We show that real and imagined movement can be told apart in people the model has never seen, at 80.25% with a 14-channel consumer headset. The result holds under person-wise cross-validation, leave-one-subject-out validation (80.25%) and nested cross-validation (79.41%).
3. **Classification of the same recordings at several levels of detail.** Every trial is classified in three ways: action or imagery (2 classes), the body part that was moved or imagined (6 classes), and the exact task, meaning body part and condition together (12 classes). The side of imagined movements, left or right, is tested as a separate two-class problem. Because all of these use the same trials, they show directly how accuracy falls as the question becomes more specific: 80.25% for action versus imagery, 26.68% for body part (chance 16.67%), 21.43% for the twelve

tasks (chance 8.33%), and 52.28% for the side of imagined movement, which is not significantly above chance.

4. **A controlled comparison of feature representations and classifiers under one subject-independent protocol.** On identical folds we compare handcrafted, spectrogram, scalogram, connectivity, CSP and Riemannian features, and classical, ensemble and deep-learning models. The Morlet scalogram is consistently more informative than the STFT spectrogram, adding CSP features to each classifier changes accuracy by less than two points, and deep networks (EEGNet, CNN–LSTM) do not outperform feature-based models on a dataset of this size.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the protocol, data acquisition, preprocessing, feature extraction, classifiers and evaluation. Section 4 presents and discusses the results, and Section 5 concludes the paper.

2 Literature Review

(to be added)

3 Proposed Methodology

(to be added)

4 Results and Discussion

(to be added)

5 Conclusion

(to be added)

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